INCH-POUND

MIL-S-24187/4B(SH)
22 March 1991

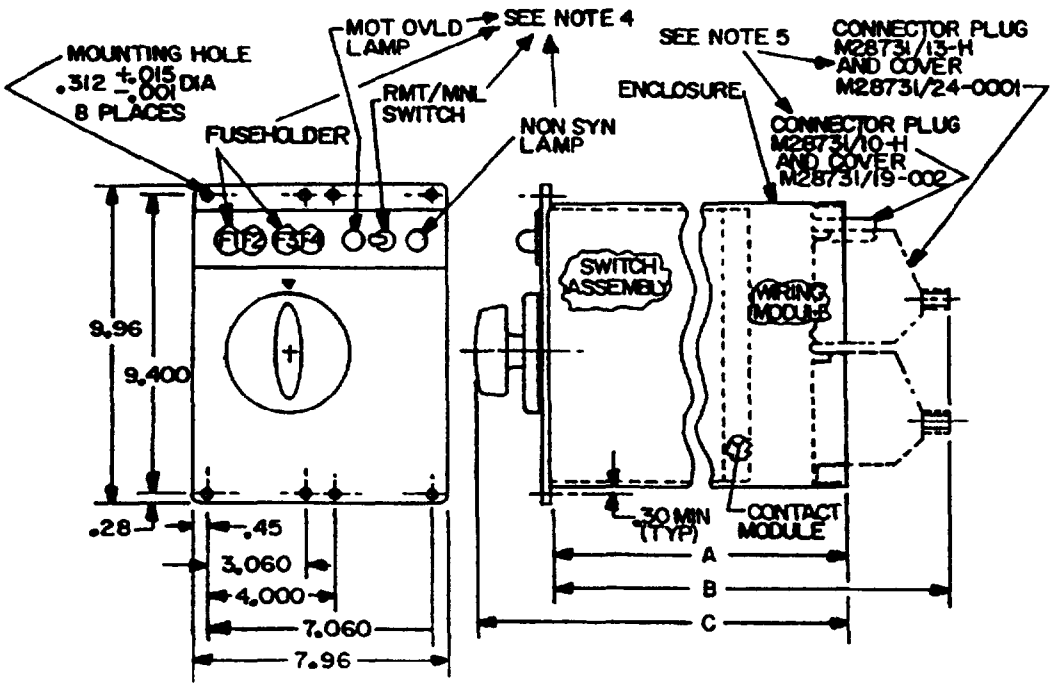
SUPERSEDING
MIL-S-24187/4A(SH)
9 September 1983

MILITARY SPECIFICATION SHEET

SWITCH ASSEMBLIES, LINEAR MOVEMENT, MANUAL AND REMOTE OPERATION, STYLE 3DLS

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-24187.



INCHES	MM
.001	0.03
.005	0.13
.02	0.51
.03	0.76
.12	3.05
.28	7.11
.30	7.62
.312	7.33
.45	11.43
3.060	77.72
4.000	101.63
7.060	179.32
7.96	202.18
9.40	38.76
9.96	252.98
14.00	355.60
16.67	423.42
18.00	457.20
18.50	469.90
20.67	525.02
22.50	571.50

FIGURE 1 *Style DLS switch assembly*

AMSC N/A FSC 5930

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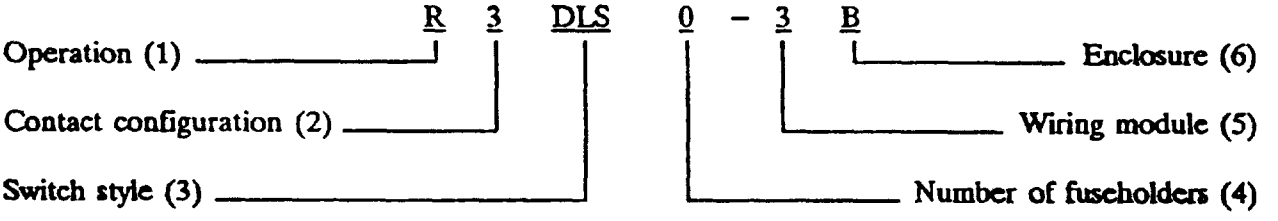
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DIMENSION	ENCLOSURE	
	A (long)	B (short)
A	18.00 ± .02	14.00 ± .03
B	22.50 ± .12	18.50 ± .12
C	20.67 ± .12	16.67 ± .12

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are plus or minus 0.02 (0.51 mm) for two-place decimals and plus or minus 0.005 (0.13 mm) for three-place decimals.
3. Connector cover, piece M28731/24-0001, may be rotated 180 degrees to accept cable from bottom.
4. Supplied only on types RDLS.
5. Supplied only when specified in contract or order.
6. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. *Style DLS switch assembly – Continued.*

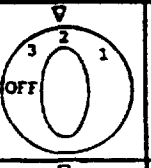
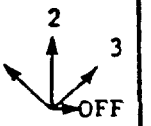
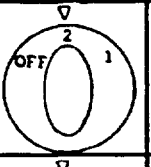
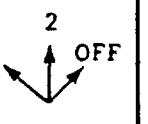
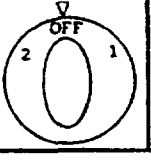
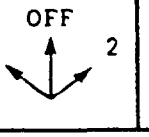
Type designation:



(1) Switch assemblies shall be remote operated with manual override, designated R, or manually operated, designated M.
(2) Contact configuration shall be type 3 (see figure 2).
(3) Switch style shall be designated by the letters DLS.
(4) There shall be no provisions for additional fuseholders.
(5) Wiring module circuits shall be indicated by a single digit (1, 2 or 3) as specified to accomplish the switching actions indicated in table I.
(6) Enclosed switch assemblies shall be indicated by a single letter (A), for long enclosure 18 inches (457.2 mm) in accordance with Drawing 803-4680149, or (B) for short enclosure 14 inches (355.6 mm) in accordance with Drawing 815-5727186

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TABLE I. *Switch selection.*

Type designation						Application data						
Switch operation	Contact configuration	Switch style	Additional fuseholders	Wiring module	Enclosure if required	Switching action[1]	Stop position	Dial position[2]	Handle position[3]	Wired contacts[4] Loading switching[6]	To ship wiring connection[5]	Unwired contact[4]
R or M	3	DLS	0	−1	A (LONG) & B (SHORT)[7]	1-2-3-OFF	2&6			( )C LOAD 3 ( )B LOAD 2 ( )A LOAD 1 ( )X INPUT	JH, JK JG, JJ JD, JF JC, JE	( )D OFF
R or M	3	DLS	0	−2	A (LONG) & B (SHORT)[7]	1-2-OFF	2&7			( )B LOAD 2 ( )A LOAD 1 ( )X LOAD	JG, JJ JD, JF JC, JE	( )D ( )C OFF
R or M	3	DLS	0	−3	A (LONG) & B (SHORT)[7]	1-OFF-2	2&7			( )C LOAD 2 ( )A LOAD 1 ( )X INPUT	JH, JK JD, JF JC, JE	( )D ( )B OFF

Example: Type designation for a 1-OFF-2 switch assembly with short enclosure R3DLSO-3B The same switch assembly without enclosure and wiring module is R3DLSO.

[1]Stops shall be adjustable to limit switch handle rotation to the positions indicated or to any consecutive positions within the maximum rotation indicated.

[2]Reference position of a DLS switch is considered to be that in which the flat on the handle shaft is up The mark ▽ and the dial positions shown are used to designate the reference position.

[3]Normal position of a DLS switch is the position which coincides with normal circuit conditions in any application. Handle shall be adjustable to point upward in the normal position or to any specified position, in 45 degree increments from upward. The handle and rotary description plate rotate.

[4]See figures 2 and 3 for complete description of contact identification, table V for load switch wiring.

[5]See figure 5 for connector identification.

[6]See table VI for circuit monitoring wiring, table VI and figure 6 for control circuit wiring.

[7]See type designation note (6) for "A" or "B" enclosure.

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REQUIREMENTS:

Construction: Style DLS switch parts, modules and assemblies shall be constructed in accordance with Drawing 803-4680149.

Characteristics: Characteristics shown in table II shall apply to the style 3DLS switch assemblies.

TABLE II. *Characteristics.*

Characteristics	Value
Total number of poles available	210
Wired load switching poles	186
Wired circuit monitoring poles	8
Wired shielding and ground poles	6
Number of positions	2 to 4
Break-before-make	Yes
Make-before-break	No
Electrical rating and endurance loading:	
2 Amperes, 15 V, dc	20,000 Operations
25 Microamperes, 1.5 V, dc	20,000 Operations
Remote operation:	
Transfer time:	
Minimum	0.010 second
Maximum	0.035 second
Actuating time:	
1 Position change	0.25 second
3 Position change (one direction)	1.10 second
Current drain:	
At 115 V, 3Ø, 60 Hz	1 Ampere rms per phase
Contact resistance:	
At 2 amperes	0.025 ohm
At 2.5 microamperes	0.05 ohm
Total circuit resistance:	
At 2 amperes	0.065 ohm
At 25 microamperes	0.100 ohm
Torque – shaft rotation maximum – Inital	90 pound-inches
After shock	100 pound-inches
– stop screw (strength)	250 pound-inches
Maximum weight.	
Enclosed with wiring module	40 pounds
Without enclosure and wiring module	30 pounds
Temperature and humidity (for rotational torque and endurance tests)	
Minimum	Plus 40 °F (plus 4 °C)
Maximum	Plus 149 °F (plus 65 °C) and 40 to 50 percent rh
Shock	MIL-S-901, grade A, class I, type C
Vibration	MIL-STD-167-1, type I, except omit variable frequency
Temperature rise	30 °C

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Switch assembly modules: DLS manually operated switch assemblies shall consist of the following modules as detailed on Drawing 803-4680149:

Switch
- Front module
- Mechanism module
- Switch contact module

Wiring module

Enclosure

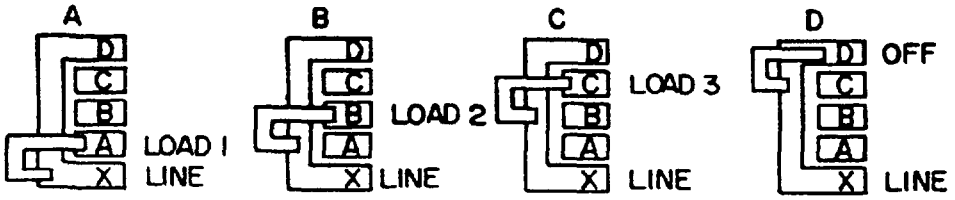
Remote operated, style DLS switch assemblies shall have, in addition to the above, the

Remote control module/mechanism module (associated circuitry)

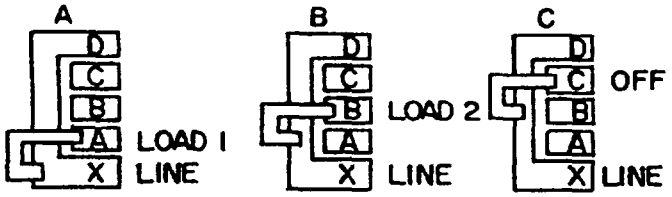
Control circuit module

Switch function: Style DLS switch assemblies shall provide the number of poles indicated, each consisting of five stationary contacts X, A, B, C, and D and a movable contact arranged as shown on figures 2 and 3. A stop (screw) shall prevent transfer of the movable contact into position X. Additional stops shall be provided which, when inserted as specified in table III, will limit movement to two, three or four consecutive positions. Stops shall be as detailed on Drawing 803-4680149 and shall be accessible from the front of the switch assembly, after removal of the handle and rotary description plate (see 6.2 of MIL-S-24187).

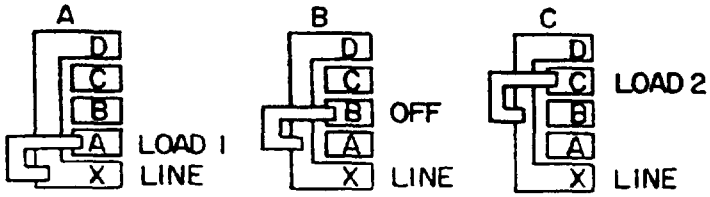
-1 Wiring module (see table I)



-2 Wiring module (see table I)



-3 Wiring module (see table I)



Example: For a 1-OFF-2 operation, the stops shall be located to prevent the switch from going to the X and D positions. Load 1 shall be connected to contact A, contact B is not connected and load 2 shall be connected to contact C.

FIGURE 2. *Schematic typical operation of 1 pole*

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TABLE III. *Stop positions versus switching action.*

Stop position	Switch positions accessible	Switching action[1]
2 and 6	A-B-C-D	1-2-3-OFF
2 and 7	A-B-C	1-2-3 1-2-OFF 1-OFF-2 OFF-1-2
2 and 8	A-B	1-2 ON-OFF OFF-ON

[1]With stops located as specified, a switch assembly shall provide two or three active positions, any of which can be used as an OFF position by omitting external connections to the terminals in the desired OFF position.

Contact designation and marking: Each stationary switching contact shall be identified by a number, two letters and a second number. The first number shall identify the switch section in which the contact is located. The first letter, A, B, C, D, E, F, or G shall identify the pole group in which a contact is located and the second letter shall be assigned as shown on figure 2. The second number shall be that of the column in which a contact is located (see figure 3). Each contact in the switch contact module connector plug shall be assigned the same identification as the contact in the switch contact module to which it connects. Contact D is common with contact X in all poles. Contacts shall be installed as shown on figure 3, circles signify omitted contacts, phantom lines indicate mating header plate.

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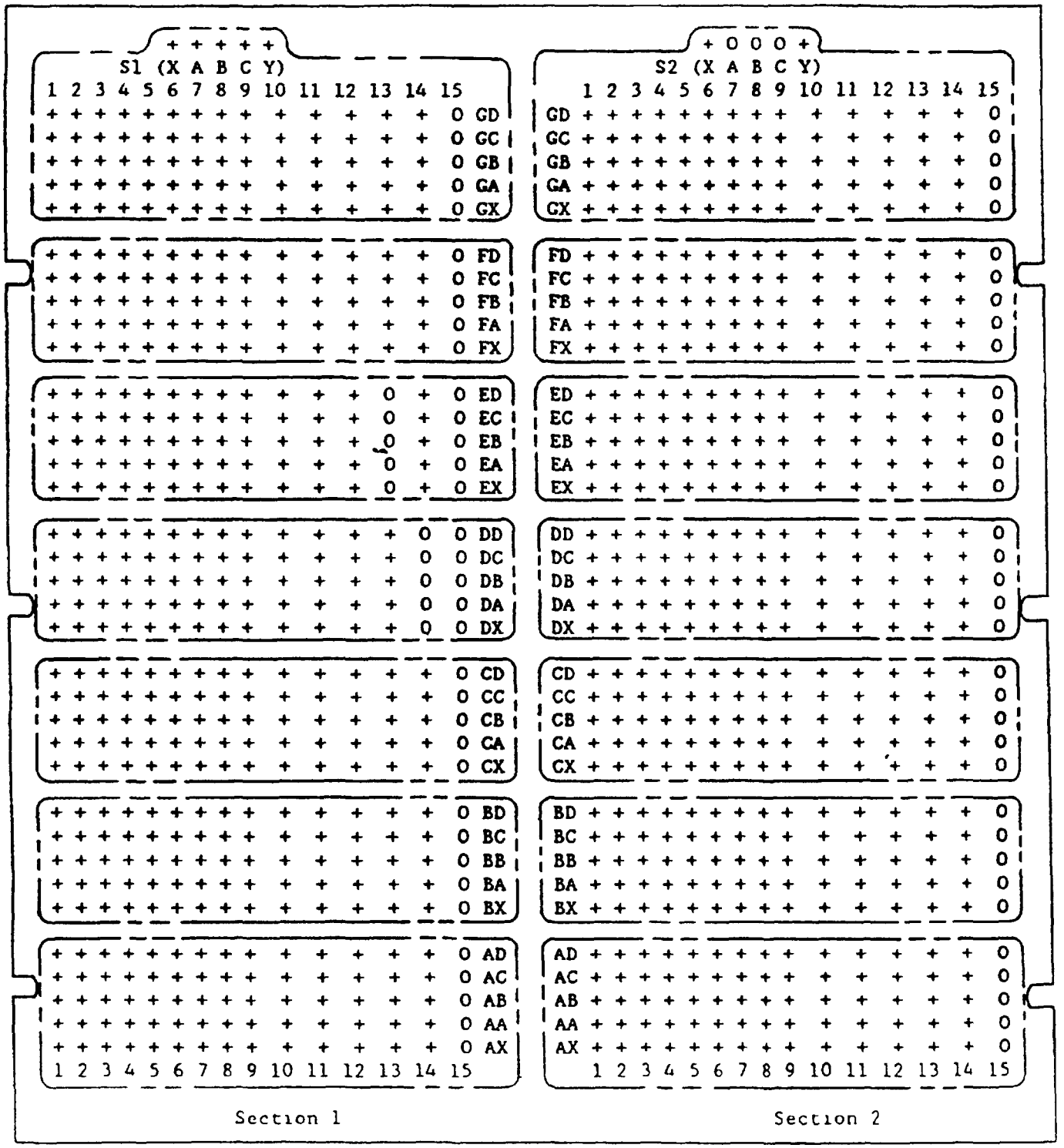
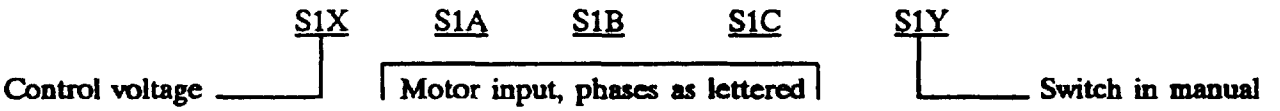


FIGURE 3 *Contact designation and marking rear view*

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Remote operation: Connections to the switch module for remote operation shall be made to section 1 contacts as follows:



Remotely operated switch assembly shall move to the positions shown in table IV when the specified control voltage is applied.

TABLE IV. *Control voltage for remote operation.*

Control voltage	Closed switch contacts			XFMR tap[1]
	−1	−1	−3	
	1-2-3-OFF	1-2-OFF	1-OFF-2	
0	X-A	X-A	X-A	A
16 4	X-B	X-B	X-B (off)	B
32.8	X-C	X-C (off)	X-C	C
49.2	X-D	Not used	Not used	D

[1]See figure 3 of MIL-S-24187 for typical test circuit for remote operated switch assemblies.

Enclosed and module: Enclosure shall be furnished with the wiring module to connect the wiring witch contact module to the connectors at the rear of the enclosure (see figure 5). The general arrangement of the wiring shall be as shown on figure 4. The interconnecting wiring shall be as shown in tables V and VI.

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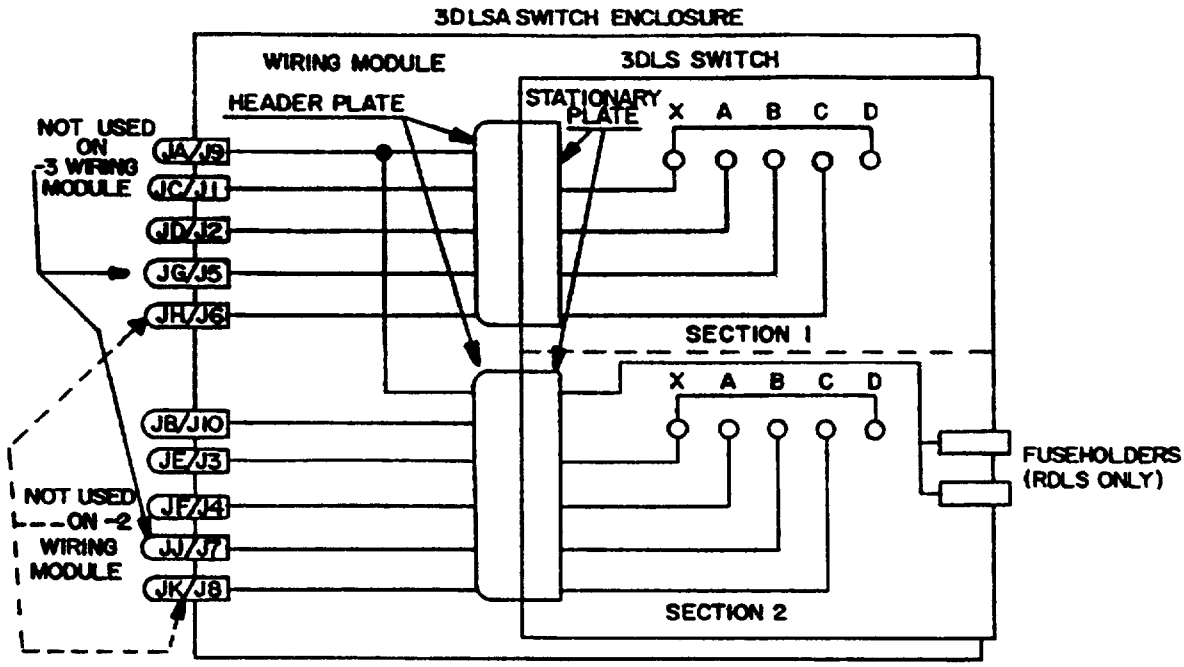


FIGURE 4. *Interconnecting wiring arrangement. (-1 wiring module)*

Wiring module connections:

Unless otherwise specified in the contract or order, wiring modules shall provide wiring for 186 load switching poles as specified in table V, plus eight circuit monitoring poles as specified in table VI. Wired contacts in each of the 194 poles shall be as specified in table I. Module wiring for switch control and indicating circuits shall be in accordance with table VI and figure 6. Wire shall be type B or E-20, 19 strand, of MIL-W-16878. The color coding shall be: X to JC/J1 and JE/J3 black, A to JD/J2 and JF/J4 white, B to JG/J5 and JJ/J7 red, and C to JH/J6 and JK/J8 blue. For shield circuits pole F13 and G14 in both sections 1 and 2 (contacts X, A, B, and C) shall be wired to pins A and B, respectively for each 120 pin connector using green wire. For other than switching circuits pole D13 in both sections 1 and 2 (contacts X, A, B, and C) shall be wired to pin DH of each 120 pin connector using gray wire.

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TABLE V. *Wiring module connections, 120 pin connectors.*

Switch contact module contact															
	Column														
Pole[1]	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
G		L	M	N	Y	P	Z	AA	R	T	S	AM	AC	B	
F	V	AE	W	AF	X	AH	AK	AJ	AB	AL	AX	AW	A		
E	AP	AR	AZ	BA	AS	AT	BC	BB	AU	AV	BD	BE			
D	BJ	BX	BT	BS	BL	BK	BM	BN	BU	BV	BP	BF	DH		
C	CE	CP	BZ	BY	CF	CH	CJ	CK	CA	CB	CL	CM	CX	CC	
B	CR	CZ	DA	DB	CS	CT	DC	DN	CU	CV	DD	DE	DF	CW	
A	DJ	DU	DV	DK	DL	DW	DM	DX	DY	DZ	DR	DP	EA	DS	
120 Pin connector contact															

[1]Each pole contains contacts X, A, B, C, and D (see figure 4)

Example: Pole E5 contact X of section 1 shall be wired to pin AS of connector JC/J1 using a black wire.
Pole E5 contact A of section 1 shall be wired to pin AS of connector JD/J2 using a white wire.
Pole E5 contact B of section 1 shall be wired to pin AS of connector JG/J5 using a red wire.
Pole E5 contact C of section 1 shall be wired to pin AS of connector JH/J6 using a blue wire.
Pole E5 contact D of section 1 shall be unwired and common to contact X.
Pole E5 contact X of section 2 shall be wired to pin AS of connector JE/J3 using a black wire.
Pole E5 contact A of section 2 shall be wired to pin AS of connector JF/J4 using a white wire.
Pole E5 contact B of section 2 shall be wired to pin AS of connector JJ/J7 using a red wire.
Pole E5 contact C of section 2 shall be wired to pin AS of connector JK/J8 using a blue wire.
Pole E5 contact D of section 2 shall be unwired and common to contact X.

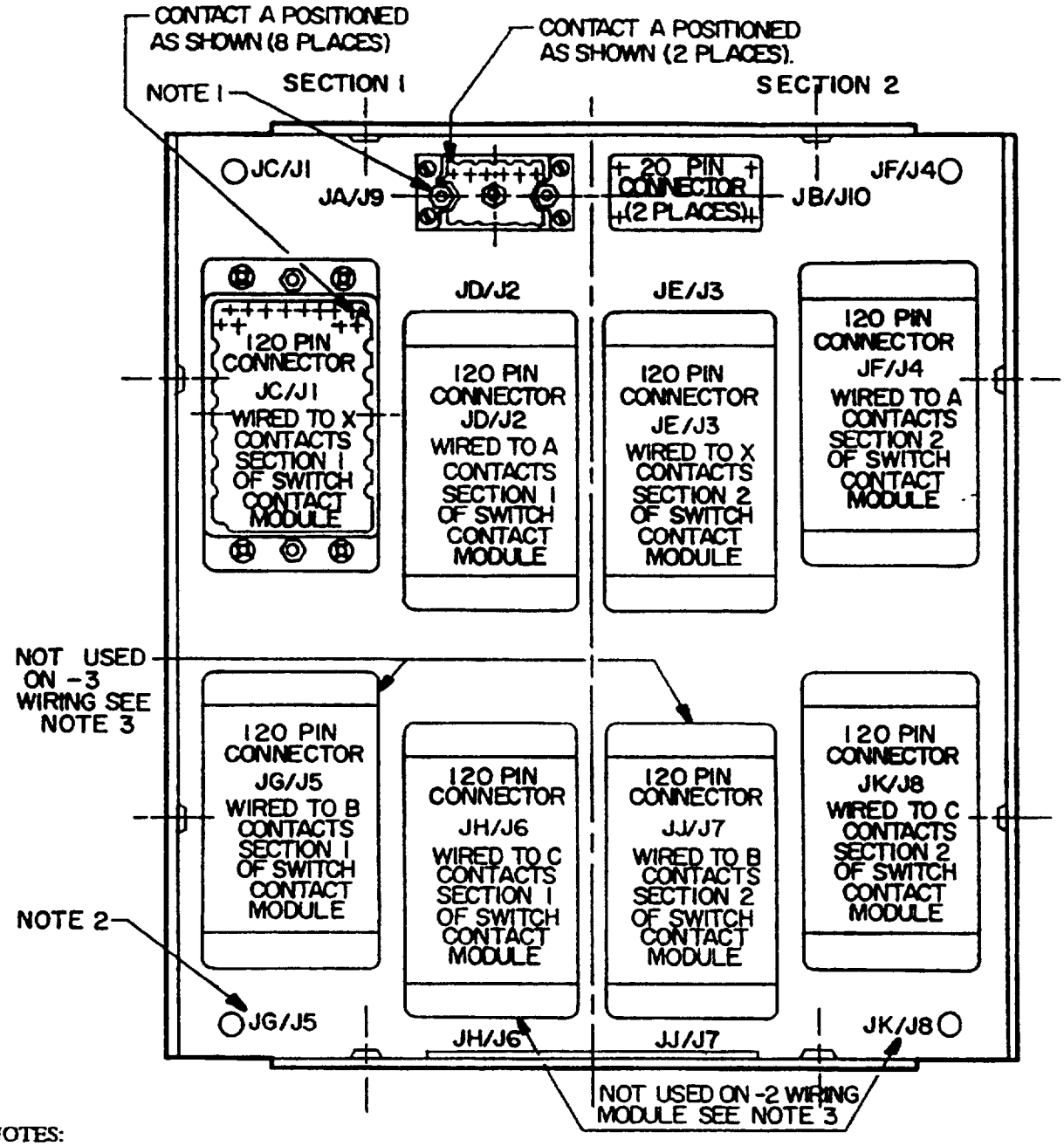
Each 20 pin connector shall be wired to the specified switch contacts as shown in table VI using gray wire. Switch module contacts 1A, 1B, 1C, 1X, 1Y, 2X, and 2Y are for the remote circuit. Remaining contacts are for circuit monitoring.

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TABLE VI *Wiring module connections, 20 pin connectors.*

20 Pin connector contact	From JA/J9 to switch control module	From JB/J10 to switch control module
A	1A	2GX1
B	1B	2GA1
C	1C	2GB1
D	1X	2GC1
E	1Y	2FX14
F	1GX1	2FA14
H	1GA1	2FB14
J	1GB1	2FC14
K	1GC1	2EX14
L	1FX14	2EA14
M	1FA14	2EB14
N	1FB14	2EC14
P	1FC14	2EX13
R	1EX14	2EA13
S	1EA14	2EB13
T	1EB14	2EC13
U	1EC14	2DX14
V	2X	2DA14
W	2Y	2DB14
X		2DC14

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NOTES:

1. On connector JA/J9, polarizing pins and sockets A and B shall be in position 1. On connector JB/J10, polarizing pins and socket A shall be in position 1 and pin and socket B shall be in position 4 (see MIL-C-28731/9 and MIL-C-28731/10)
2. Characters 0.19 inch high (4.83 mm), black epoxy ink. Location shall be visible with connector covers in place (see figure 1)
3. Cover plates shall be used in place of connectors in those applications not needing connectors.

FIGURE 5. *Enclosed switch assembly connector arrangement, rear view.*

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Remote control circuit: Control circuit for remote operation shall be in accordance with figure 6.

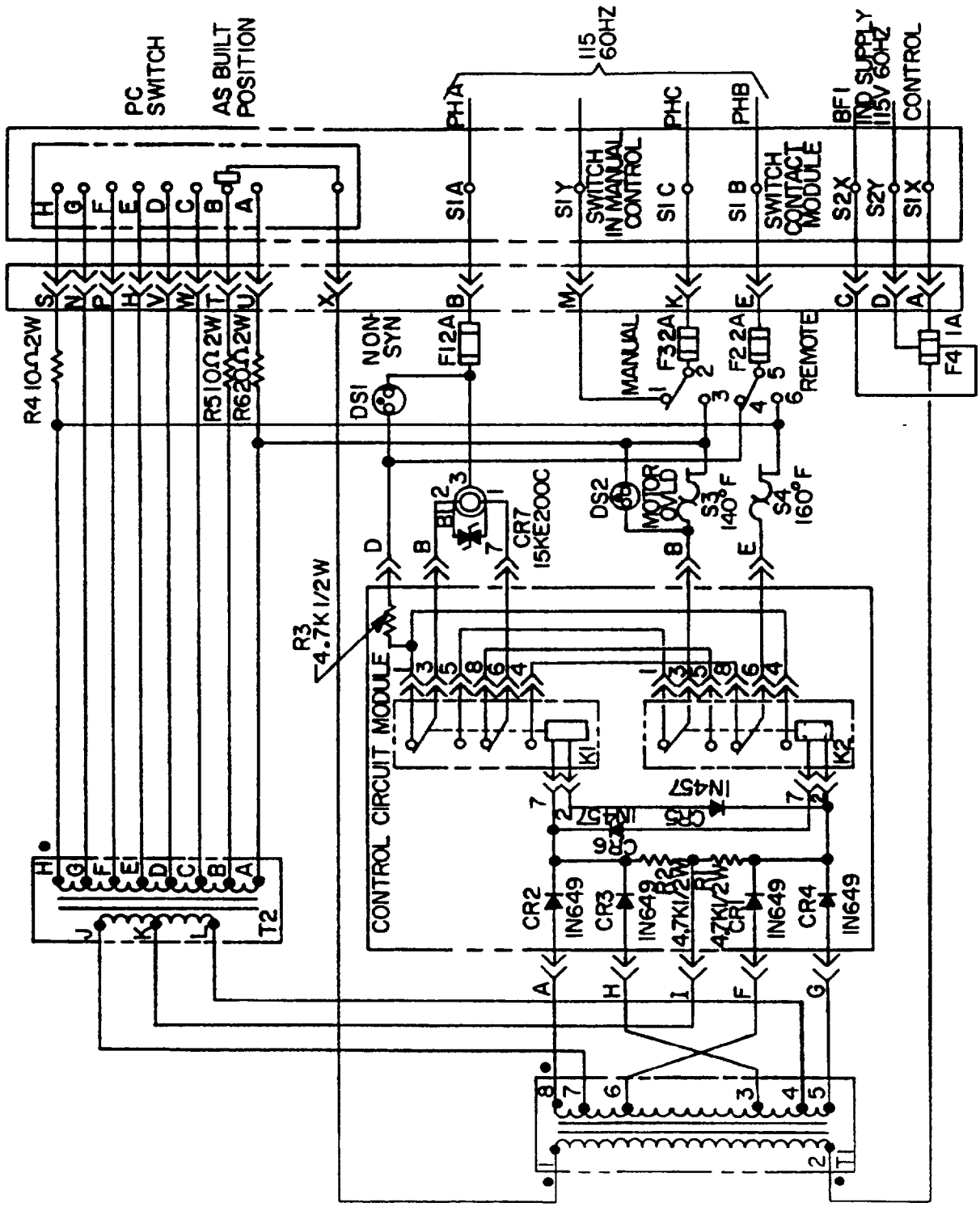


FIGURE 6. *Remote control circuit.*

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Repair parts: Repair parts shall be as specified in table VII.

TABLE VII. *Repair parts.*

Drawing 803-5476870 piece No.		Description	Quantities required for quantity of switches installed					
			1–3	4–12	13–30	31–60	61–120	121 or more
Manual	Remote							
	1	Switch, type R3DLSO-1A	1	1	1	1	2	2
	214	Switch, type R3DLSO-2A	1	1	1	1	2	2
	401	Switch, type R3DLSO-3A	1	1	1	1	2	2
	301	Switch, type M3DLSO-3A	1	1	1	1	2	2
	501	Switch, type M3DLSO-1A	1	1	1	1	2	2
	601	Switch, type M3DLSO-2A	1	1	1	1	2	2
		815-5727186						
	1	Switch, type R3DLSO-1B						
	2	Switch, type R3DLSO-2B						
	3	Switch, type R3DLSO-3B						
4		Switch, type M3DLSO-1B						
5		Switch, type M3DLSO-2B						
6		Switch, type M3DLSO-3B						
158	158	Stop screw	1	1	1	2	2	3
32	32	Detent spring	1	1	1	2	3	6
	87	Toggle switch	1	1	1	2	3	4
	85	Indicator light	1	1	1	2	3	5
	86	Indictor light	1	1	1	2	3	5
		Lamp (M15098/11-001)	Note 2					
	10	Control circuit module	1	1	1	2	2	3
	93	Relay	1	2	3	Note 3		
	160	Washer, insul.	2	2	2	4	6	8
	159	Spacer, insul.	2	2	2	4	6	8
	81	Fuseholder	1	1	1	2	3	4
		Fuse carrier (1A0981)	1	1	2	3	4	5
	82	Fuse, 2.0 ampere	Note 4					
	83	Fuse, 1.0 ampere	Note 4					

NOTES

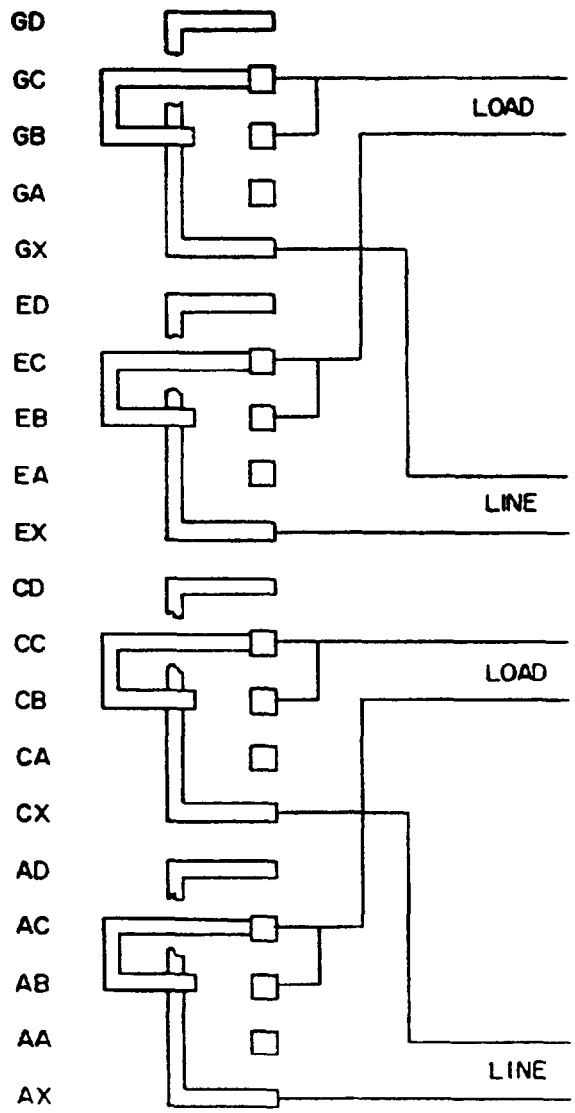
1. Refer to Drawing 803-4680149 for piece number(s).
2. One hundred percent repair lamps shall be provided.
3. Supply one for each 10 or fraction of 10 installed.
4. One hundred fifty percent repair fuses shall be supplied Fuses shall be supplied in multiples of five, in standard commercial packages, with a minimum quantity of 15
5. Connector plugs, piece number 210 and 211, are not to be supplied

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Qualification samples: Two remote operated switch assemblies type R3DLS01A shall be submitted for qualification inspection.

Tests:

Loading of switches: Loading of switches for endurance and temperature rise tests for qualification and quality conformance, group C testing, shall be as shown on figure 7 and table VIII.



NOTE Sections 1 and 2, row 1, shall be loaded at 2 amperes resistive at 15 volts dc. Sections 1 and 2, row 14, shall be loaded at 25 microamperes resistive at 1.5 volts dc.

FIGURE 7. *Test loading*

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Temperature rise: Contacts to be monitored during temperature rise tests are specified in table VIII.

TABLE VIII. *Temperature measurements.*

Section	Row	Contacts
1	1	GX, GB, CX, CB
2	1	EX, EB

Revision letters are not used to denote changes due to extensiveness of the changes.

Preparing activity:
Navy – SH
(Project 5930-N681-05)

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